Candidate's Seat no: ____________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2014

EE308/EE308.01 High Voltage Engineering

Date: 12.05.2014, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Define the following terms: [07]

- Breakdown Voltage
- Dielectric Strength
- Impulse Wave
- Time Lag
- Arc Voltage
- Corona Discharge
- Flashover

Q - 2 (a) Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in Townsend's discharge? [07]

(b) In an experiment, in a certain gas it was found that the steady state current is 5.5×10^{-8} A at 8 kV at a distance of 0.4 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.1 cm results in a current of 5.5×10^{-9} A. Calculate Townsend's primary and secondary ionization coefficient. [07]

OR

Q - 2 (a) Explain the Streamer theory of breakdown in air at atmospheric pressure. [07]

(b) Explain post breakdown phenomenon in detail. [07]

Q - 3 (a) Explain electromechanical breakdown phenomenon in case of solid dielectrics. [06]

(b) Explain the characteristics of liquid insulations. [04]

(c) List out the various breakdown phenomenon observed in solid dielectrics. [04]

OR

Q - 3 (a) Explain the phenomenon 'treeing and tracking' in solid insulating materials under electrical stress. How does it lead to breakdown? [07]

(b) Explain the various classifications of liquid dielectrics. [07]

SECTION – II

Q - 4 Explain how a sphere gap can be used to measure the peak value of voltages? Also discuss the various factors influences such voltage measurement. [07]

Q - 5 (a) Explain the method of cascade transformers used in generation of high voltages. [07]

(b) Explain tesla coil used to generate high frequency high voltage ac with suitable circuit diagram. [07]

OR

Q - 5 (a) A 12 – stage impulse generator has 0.126 μF capacitors. The wave front and the wave tail resistances connected are 800 ohms and 5000 ohms respectively. If the load capacitor is 1000 pF, find the front and tail times of the impulse wave produced. [07]

(b) A generating voltmeter has to be designed so that it can have range from 20 to 200 kV dc. If indicating meter reads a minimum current of 2 μA and maximum current of 25 μA, find (1) the capacitance of generating voltmeter and (2) the value of current indicated by meter when voltage is 200 kV. Assume that the driving motor has a synchronous speed of 1500 rpm. [07]

Q – 6 (a) Explain the layout of short circuit plant used for the testing of circuit breakers. [07]

(b) Explain impulse testing of transformer. [07]

OR

(a) With neat sketch explain High Voltage Schering Bridge used for the measurement of unknown capacitance and dissipation factor. [07]

(b) With neat sketch explain the phenomenon of partial discharge and derive the equation for the energy associated in a single discharge. [07]

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